
Label Swapping for Imbalanced Music Datasets

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1 NOTES

$$\rho[i_{MA}] = \max(\tanh(\mathcal{Z}[i_{MA}] - t_{flip}), 0)$$

- $\mathcal{Z}[i_{MA}]$: Z-score for each majority-class sample. High Z means that sample looks like minority class.
- t_{flip} : Threshold that decides how large Z must be before a sample gets any change of being flipped.
- $\mathcal{Z}[i_{MA}] \leq t_{flip} \rightarrow \tanh(\mathcal{Z}[i_{MA}] - t_{flip}) \leq 0 \rightarrow \text{flip-rate} = 0$
- $\mathcal{Z}[i_{MA}] > t_{flip} \rightarrow \text{flip-rate grows as Z increases}$

Project

While giving basic knowledge about the paper is fine and sometimes necessary to be able to expand on it, the report should be mostly focused on what your group did in the project. As a guideline, the report can contain the following sections:

- **Abstract (mandatory):** Give an overview of what you have done in the project with the key results and findings of your work. Should be no more than 300 words.
- **Introduction (mandatory):** Describe the problem, the approach of the paper, the experiments, and the results. At the high-level talk about what you worked on in your project and why it is important. Then give an overview of your results.
- **Code repository (mandatory):** link to the KTH repository containing the code corresponding to all implementations and experiments of the report
- **Related Work (mandatory):** Discuss the published work related to your project paper, the types of experiments you do and the additional method that you have added to this work or you have compared this paper with (if any).
- **Methods (mandatory):** Describe the original paper's method to the extent that you would need to make your report and findings understandable. Otherwise, here you can describe other methods that you compare with or other methods that you apply on top of the main method of the paper. Here, you also try to justify any methodical modification or incremental changes that you have added to the original paper. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.
- **Data (mandatory):** Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, filtering, or other special treatment to use this data in your project?
- **Experiments and findings (mandatory):** Describe and present the experiments that you performed in detail and what is the reason for those experiments. Where applicable define evaluation metrics that you used. Discuss the results that you got. You should include graphs, tables, or other figures to illustrate your experimental results. This is the most important part of your report, be thorough, clear and detailed. Does the reproduction show a significant deviation from those results reported in the original paper? What could be the reason? What extra experiments can you think of to substantiate those speculated reasons? Do the conclusions, observations, statements, and theories of the original paper hold in the new experiments or datasets that you have tried? To what extent? Discuss the similarities and discrepancies and the possible reasonings.
- **Challenges (mandatory):** Challenges you faced when reproducing and/or reimplementing the paper and conducting the experiments. Were all details in the paper? Or did you have to look in the authors code or contact authors to find about some details (you are encouraged to contact the authors)? Was parts of the code quite hard to get them to work as intended? Did you have to optimize and tune several hyperparameters? Which ones? Did the framework you used make the implementation difficult in some ways?
- **Conclusion (mandatory):** Summarize your key results - what have you learned? What points do you think one should consider when using the approach of the paper you chose for your project? What can go wrong when reimplementing based on paper and why? A summary of similarities and discrepancies that you found when conducting the experiments. Suggest ideas for future extensions or new applications of your ideas.
- **Ethical consideration, societal impact, alignment with UN SDG targets (mandatory):** Think and research! Are there any ethical considerations for the original paper, its problem or method, its way of conducting experiments? How about the experiments you did? What societal impact can you imagine about the original paper and its contributions and results? How about your project report? How do you think this paper can push or inhibit the UN SDG targets?
- **Limitations (mandatory):** what are the limitations of the paper in general and in particular its presentation, its theory, its proposed technical approach, its conducted experiments, its discussions, and its conclusions?
- **Self Assessment (mandatory):** Use the [Project Grading Guideline](#) again and assess your project along those guidelines. Argue what grade you think your project deserves and why.
- **Self-nomination for an upgrade in project grade (optional):** Do you believe that you are entitled to an upgrade in your final grade because you went above and beyond what you were supposed to do in one or several aspect of the final project? Here's your chance to *briefly* point out those aspects.

Abstract

Data imbalance is a pervasive issue in machine learning that often hinders classifier performance by biasing predictions towards the majority class. Label-Noise-based Rebalancing (LNR) is a novel method proposed by Hu et al. (2025) that introduces beneficial label noise to adjust biased decision boundaries without the drawbacks of traditional resampling techniques. We apply this method to the imbalanced music dataset Meter2800 for time signature detection, meaning to improve the detection of rare, irregular meters (e.g., 5/4, 7/4) using a ResNet18 architecture. Our results show that beneficial label noise yields improved performance for recognizing minority class time signatures.

2 Introduction

Data imbalance is a common issue in machine learning that hinders classifier performance, leading to biased predictions favoring the majority class. In imbalanced datasets (e.g., 90% class A, 10% class B), accuracy is maximized by favoring class A, pushing the decision boundary deep into the territory of class B. In other words, if an exam were to have 2 options, where option A is right 90% of the time, students would answer every question with option A, receiving a 90% score. There is no point learning the minority class, as it is too much work for too few points, ignoring the rare minority class, as it wants to maximize its overall accuracy score.

This results in poor performance for the minority class. Additionally, techniques such as resampling often introduce information loss or overfitting. Finding a solution is relevant in scenarios where overrepresentation of one class over another may be problematic.

Recently, Hu et al. (2025) proposed a novel approach, Label-Noise-based Rebalancing (LNR), suggesting that adding beneficial label noise has the potential to adjust biased decision boundaries and improve performance without the drawbacks of traditional resampling.

Our project means to apply this method to imbalanced music datasets, specifically time signature detection. Meter2800 is an imbalanced dataset of musical excerpts labeled with time signatures, where standard meters 4/4 and 3/4 are abundant, while irregular meters such as 5/4 and 7/4 are scarce.

Reproducing the results of ResNet18 on Meter2800 and augmenting the training process with LNR, we seek to establish a new way of detecting complex musical meters.

Code repository is available at https://github.com/Matdrox/dl_music_label_swapping

3 Related Work

3.1 Label Noise Rebalancing (LNR)

Binary classification on imbalanced datasets often results in biased decision boundaries, causing misalignment between accuracy and metrics such as F1-score. LNR addresses this by introducing asymmetric label noise, selectively flipping majority class samples to the minority class based on their similarity to minority class samples. Unlike symmetric noise, which affects all classes equally, LNR focuses on majority samples that are confusing (i.e., they have high posterior probabilities of looking like the minority class). This process pushes the decision boundary back toward the majority region, reclaiming space for the minority class without discarding any data.

3.2 Time Signature Detection

Audio Similarity Matrices (ASM) and Beat Similarity Matrices (BSM) have been used as early methods for meter detection relying on signal processing. More recently, deep learning has shown superior performance. Abimbola et al. (2024) [1] demonstrated that a ResNet18 architecture using Mel-frequency Cepstral Coefficients (MFCCs) as input outperformed traditional machine learning models such as SVM and KNN, as well as other deep neural networks like CNN and CRNN. However, the dataset used in the paper, Meter2800, is highly imbalanced, making it a suitable candidate for testing the limits of the LNR method.

4 Methods

The final model architecture combines ResNet18 for time signature detection with the LNR algorithm for handling data imbalance. Imperfect representation is a classical problem in music information retrieval, where some musical concepts are inherently rare. Time signatures such as 5/4 and 7/4 are uncommon in Western music, while 4/4 and 3/4 dominate. This leads to imbalanced datasets that bias classifiers toward the majority classes, resulting in poor detection of minority class time signatures.

4.1 ResNet18 for Time Signature Detection

Time signature detection is approached as a supervised image classification problem using the methodology instructed by Abimbola et al. (2024) [1]. Raw audio signals are pre-processed into visual representations and fed into a ResNet18 architecture adapted for single-channel audio features.

4.1.1 Feature Extraction

Audio files are converted into Mel-frequency Cepstral Coefficients (MFCCs), compactly representing the spectral properties of the audio signal. For each 30-second audio clip, 13 cepstral coefficients are extracted over 130 time frames. In order to maintain consistency, the input dimensions are fixed to 130×13 (time frames $\times$ frequency coefficients). Each input sample x_i is thus treated as a single-channel image tensor $x_i \in \mathbb{R}^{1 \times 130 \times 13}$.

4.1.2 Model Architecture

The Resnet18 architecture is a deep residual network consisting of 18 layers. While originally designed for three-channel RGB images (e.g., ImageNet), we adapt it for our low-resolution, single-channel audio features. MFCC inputs are processed through a series of residual blocks, letting it learn representations of the musical meters. The final layer is a fully connected layer that maps the extracted features to the output classes corresponding to the time signatures 3/4, 4/4, 5/4, and 7/4.

4.1.3 Training Configuration

Following the baseline settings, the model is trained using the Adam optimizer to minimize the cross-entropy loss function. A learning rate of 0.001 is used and a trained for 40 epochs.

4.2 LNR Algorithm

In order to enforce the model’s attention toward the minority class, LNR proposes to "lie" to the model. Instead of safely assuming the majority class, flipping the label of ambiguous samples to the minority class causes the model to pay better attention to the nuances of the minority class.

Lying randomly would confuse the model. Therefore, beneficial noise was introduced by the implementation of the LNR algorithm as follows:

- Let X be the input feature space (audio spectrograms) and $Y = \{0, 1\}$ be the binary label space, where 0 represents the majority class and 1 the minority class.
- Let f_θ be the ResNet classifier parameterized by weights θ .
- $\hat{\eta}(x_i) = P(Y = 1|X = x_i)$ denotes the estimated posterior probability (confidence score) that sample x_i belongs to the minority class.
- $I_{MA} = \{i|y_i = 0\}$ is the set of indices of majority class samples in the current batch.

The LNR algorithm identifies confusing majority samples based on their posterior probabilities. Calculating the mean μ and standard deviation σ of these probabilities after passing them through the classifier’s softmax layer, the standardized Z-score Z_i for each majority sample is computed as:

$$Z_i = \frac{\hat{\eta}(x_i) - \mu}{\sigma}$$

Flip-rate probability $\rho(i)$ is computed using a hyperbolic tangent function and a scalar threshold hyperparameter t_{flip} :

$$\rho(x_i) = \max(\tanh(Z_i - t_{flip}), 0)$$

Labels are flipped stochastically based on $\rho(x_i)$, as only majority samples with sufficiently high confusion (i.e., $Z_i > t_{flip}$) should have a non-zero chance of being flipped to the minority class. The algorithm is given in the pseudocode below:

Algorithm 1 Flip labels in binary classification with noise model $\rho(x)$

Require: Feature X and labels Y , Classifier C_f

Require: Threshold t_{flip}

$Ind_{MA} \leftarrow \text{index}(Y = 0)$

$\hat{\eta}, Z, \rho \leftarrow \text{zeroVector}(\text{size} = N)$

for $i \in 1 \rightarrow \text{length}(Y)$ **do**

$\hat{\eta}[i] \leftarrow C_f(X[i])$

end for

$\mu \leftarrow \text{mean}(\hat{\eta}[Ind_{MA}])$

$\sigma \leftarrow \text{std}(\hat{\eta}[Ind_{MA}])$

for each i_{MA} **in** Ind_{MA} **do**

$Z[i_{MA}] \leftarrow \frac{\hat{\eta}[i_{MA}] - \mu}{\sigma}$

$\rho[i_{MA}] \leftarrow \max(\tanh(Z[i_{MA}] - t_{flip}), 0)$

$U \leftarrow \text{Bernoulli}(\rho[i_{MA}])$

if $U == 1$ **then**

$Y[i_{MA}] \leftarrow 1$

end if

end for

return Y

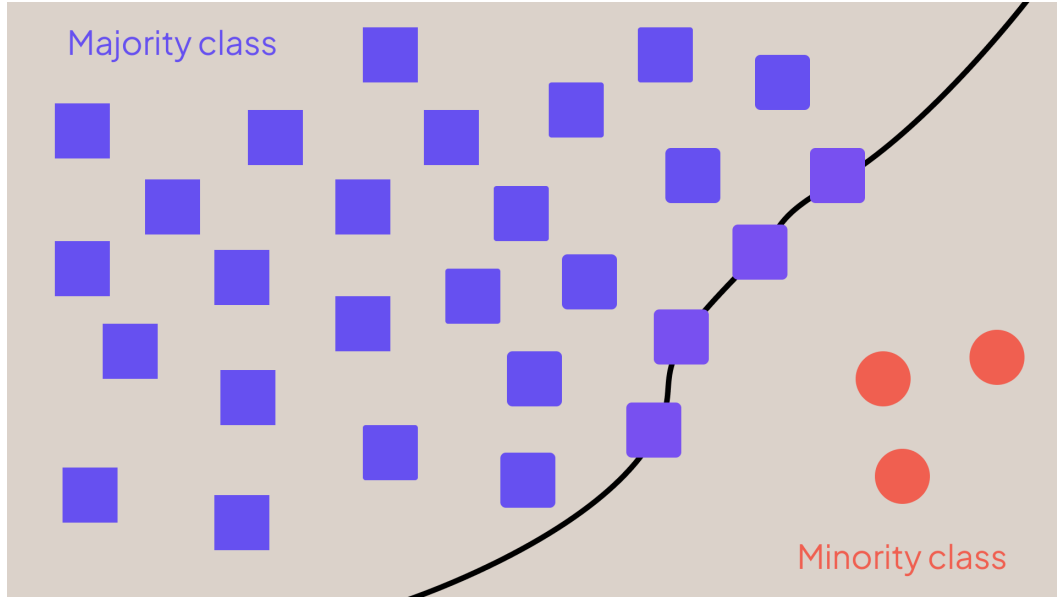


Figure 1: Representation of majority and minority class samples in feature space. *KANSKE TAR BORT*

5 Data

Three datasets are used in this project.

5.1 KEEL

For binary classification experiments replicating the original LNR paper, the same KEEL datasets are used. KEEL is a repository of 99 binary imbalanced datasets. From these, 32 specific datasets were selected by Hu et al. (2025) for their experiments, which we also use for replication.

Their chosen datasets match the following criteria:

- a) Baseline already achieves an F1-score over 90%, or all methods have an F1-score lower than 10%.
- b) Datasets with insufficient minority class sample size in the test dataset (less than 10).

5.2 CIFAR-10

Multiclass reproduction required us to use CIFAR-10, a well-known image classification dataset consisting of 60,000 32x32 color images in 10 classes. CIFAR-100 was also used in the original paper, but it was not necessary for our reproduction.

5.3 Meter2800

Meter2800 is a time signature dataset provided by Harvard Dataverse. It contains four annotated CSV files of audio data segmented into the meter classes 3/4, 4/4, 5/4 and 7/4. The dataset also contains a total of 2800 audio files of 30 seconds each. It exhibits severe class imbalance, making it a perfect candidate for testing the LNR method:

- **Majority classes** 3/4 and 4/4 (1200 samples each).
- **Minority classes** 5/4 and 7/4 (200 samples each).

6 Experiments and Findings

7 Challenges

The LNR paper covers a novel, yet fairly approachable method for imbalanced dataset classification. Algorithms in form of pseudocode are provided, and the method is easy to implement.

However, the paper is quite broad. In an attempt to present the benefits of the proposed LNR method, experiments were conducted using many different frameworks. This requires the reader to not only understand the theory behind the proposed method, but also the details of the various frameworks used in the experiments. This makes it difficult to reproduce the results without prior knowledge of the frameworks.

Additionally, the structure of the paper makes it difficult to follow the experimental setups. The datasets used are scattered throughout the paper, and it is not always clear which datasets were used for which experiments. This requires the reader to constantly refer back to previous (or future) sections to understand the context of the experiments. Even then, some details are left out, making it difficult to fully understand the experimental setups.

8 Conclusion

9 Ethical Consideration

Time signature detection is a niche task in music information retrieval with limited ethical implications. However, data integrity and representation remain important.

The Meter2800 dataset, while useful for its intended research purpose, may not fully represent the diversity of musical styles and cultures. This could lead to biased models that perform poorly on underrepresented time signatures or genres.

Fixing imbalanced datasets is an important task for machine learning in general. Real-world cases such as breast cancer detection often involve many more negative samples than positive ones, despite the importance of correctly identifying the positive cases. In these critical applications, deliberately mislabeling majority samples to improve sensitivity ought to be done with uttermost care to avoid an unacceptable increase in false positives.

10 Limitations

Meter2800, although imbalanced and suitable for testing LNR, is relatively small with only 200 samples for the minority classes. The statistical significance of the improvement is harder to establish compared to large-scale image datasets like CIFAR-10.

11 Self Assessment

The group aimed for an A by extending the LNR methodology to a novel domain of Music Information Retrieval. We successfully implemented the algorithm, explained its theory, and reproduced the original results on binary and multiclass classification tasks. Moreover, we integrated LNR with a distinct deep learning pipeline (ResNet for Audio), doing more than simple reproduction.

12 Submission of report to DD2424

We require an electronic submission to the Canvas webpage.

Please read the instructions below carefully and follow them faithfully.

12.1 Style

Papers to be submitted to NeurIPS 2018 must be prepared according to the instructions presented here. Papers may only be up to eight pages long, including figures. Additional pages *containing only acknowledgments and/or cited references* are allowed. Papers that exceed eight pages of content (ignoring references) will not be reviewed, or in any other way considered for presentation at the conference.

The margins in 2018 are the same as since 2007, which allow for $\sim 15\%$ more words in the paper compared to earlier years.

Authors are required to use the NeurIPS L^AT_EX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for rejection.

12.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the World Wide Web at

<http://www.neurips.cc/>

The file `neurips_2018.pdf` contains these instructions and illustrates the various formatting requirements your NeurIPS paper must satisfy.

The only supported style file for NeurIPS 2018 is `neurips_2018.sty`, rewritten for L^AT_EX 2_ε. **Previous style files for L^AT_EX 2.09, Microsoft Word, and RTF are no longer supported!**

The L^AT_EX style file contains three optional arguments: `final`, which creates a camera-ready copy, `preprint`, which creates a preprint for submission to, e.g., arXiv, and `nonatbib`, which will not load the `natbib` package for you in case of package clash.

New preprint option for 2018 If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text “Preprint. Work in progress.” in the footer. This version may be distributed as you see fit. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please *do not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2018.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 13, 14, and 15 below.

13 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by $\frac{1}{2}$ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow $\frac{1}{4}$ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 15 regarding figures, tables, acknowledgments, and references.

14 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

14.1 Headings: second level

Second-level headings should be in 10-point type.

14.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

15 Citations, figures, tables, references

These instructions apply to everyone.

15.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf>

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dotso
```

produces

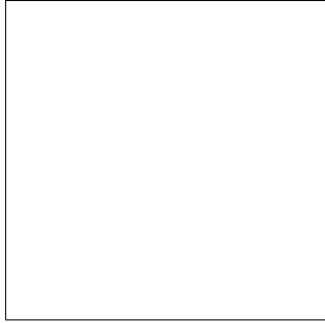


Figure 2: Sample figure caption.

Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2018` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2018}
```

As submission is double blind, refer to your own published work in the third person. That is, use “In the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form “A. Anonymous.”

15.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹ in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.²

15.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

15.4 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

¹Sample of the first footnote.

²As in this example.

Table 1: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

<https://www.ctan.org/pkg/booktabs>

This package was used to typeset Table 1.

16 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

17 Preparing PDF files

Please prepare submission files with paper size “US Letter,” and not, for example, “A4.”

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF file uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdf fonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- The IEEE has recommendations for generating PDF files whose fonts are also acceptable for NeurIPS. Please see <http://www.emfield.org/icuwb2010/downloads/IEEE-PDF-SpecV32.pdf>
- `xfig` “patterned” shapes are implemented with bitmap fonts. Use “solid” shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

```
\usepackage{amsfonts}
```

followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

```
\newcommand{\RR}{\mathbb{R}} %real numbers
\newcommand{\Nat}{\mathbb{N}} %natural numbers
\newcommand{\CC}{\mathbb{C}} %complex numbers
```

Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

17.1 Margins in $\text{\LaTeX}$

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (<http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf>)

A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the \- command when necessary.

Acknowledgments

Use unnumbered third level headings for the acknowledgments. All acknowledgments go at the end of the paper. Do not include acknowledgments in the anonymized submission, only in the final paper.

References

References follow the acknowledgments. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. **Remember that you can use more than eight pages as long as the additional pages contain *only* cited references.**

- [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.
- [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural Simulation System*. New York: TELOS/Springer-Verlag.
- [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

References

- [1] Jeremiah Abimbola, Daniel Kostrzewa, and Paweł Kasprowski. Music time signature detection using resnet18. *EURASIP journal on audio, speech, and music processing*, 2024(1):30–10, 2024.